

Interpretable Geometric Analysis of the Humerus in Fat-Suppressed Shoulder MRI

Toward localization of rotator cuff tears via foundation-model segmentation
and direction-aware geometric features

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Topics in Computational Geometry — UNAL, Colombia

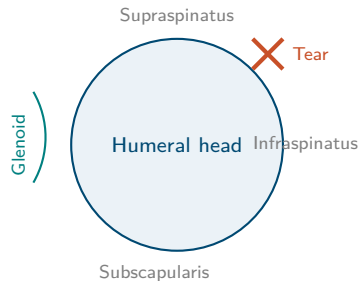
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Bridging computational geometry and musculoskeletal radiology

The clinical problem: rotator cuff tears

Why it matters

- Affects **>30% of adults over 60** — main cause of shoulder pain.
- Surgical planning depends on **where** the tear is, not only *if* there is one.
- MRI reading is expert-dependent; access to specialists is unequal (e.g. **Chocó, Colombia**).



Axial view: tears disrupt the articular contour *focally*.

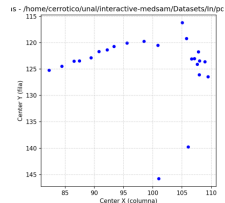
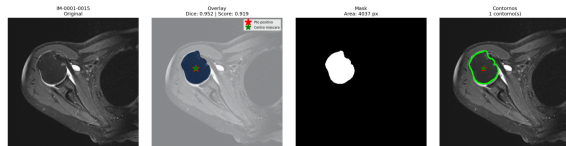
Why fat-suppressed axial MRI?

Two practical choices

- **Fat suppression:** fat appears dark, so tendons, edema and joint fluid stand out.
- **Axial plane:** the humeral head appears as *nested circles*, peaking at the equator.

What this gives us

A clean, bell-shaped *area-vs-slice* curve — the anchor of our entire pipeline.



PD-fs axial slice with humeral segmentation overlay.

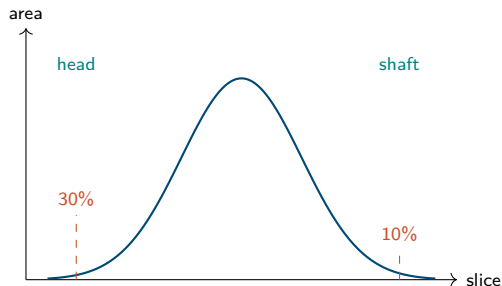
From anatomy to geometry

Two anatomical facts $\Rightarrow$ two geometric features

- Humeral head $\approx$ **sphere** (radius 23–26 mm).
- Articular contour $\approx$ **circles** that *lie on* the surface, slice by slice.

Head $\neq$ shaft

- Head closes **abruptly**; shaft tapers **gradually**.
- $\Rightarrow$ asymmetric trim: **30%** on the head side, **10%** on the shaft side.



Asymmetric trim, driven by anatomy.

Hypothesis and objective

Hypothesis

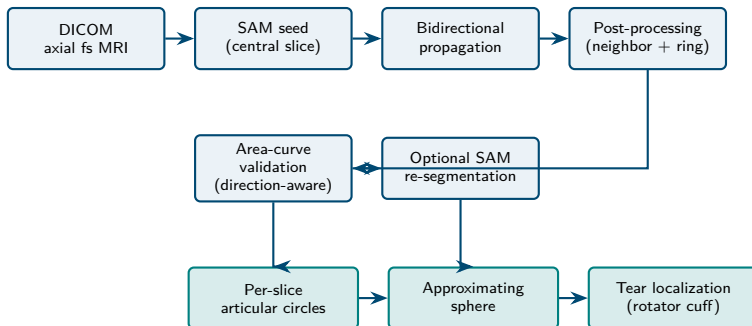
A tear leaves a **focal, geometric trace** on the articular contour. If we extract clinically meaningful geometric features, we can **localize** the lesion to a specific tendon — *without* training a new neural network.

What we built

An end-to-end pipeline that is:

- **Interpretable** — every output is a mask, a circle, or a sphere a radiologist can verify.
- **Vendor-agnostic** — works on SIEMENS, GE and Philips without retraining.
- **Anatomy-aware** — thresholds come from anatomy, not arbitrary heuristics.

Pipeline overview



Frozen SAM + classical geometry + clinical anatomy.

Every block produces **auditable** outputs: masks, CSVs, summary figures.

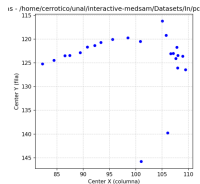
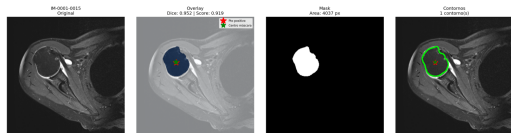
Stage 1: SAM segmentation + bidirectional propagation

Foundation model, no training

- **SAM (ViT-B)** as a frozen segmenter — no labels, no fine-tuning.
- One seed slice (one click) starts everything.
- The mask of slice i provides the prompt for slice $i \pm 1$.

Adaptive quality control

Dice threshold is **stricter near the seed, looser far from it**. If a slice fails, a negative prompt is added and the slice is re-segmented.



Per-slice segmentation: green = mask boundary, red = prompt.

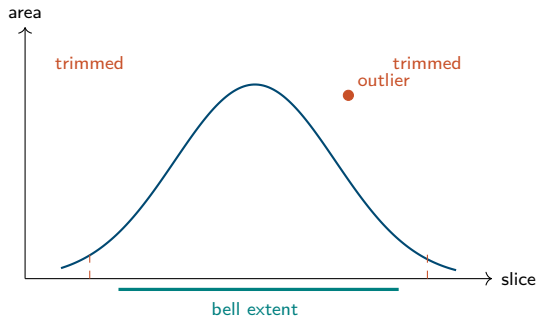
Stage 2: validating the area curve

Read direction from the DICOM header

We compute the slice normal from ImageOrientationPatient to detect **which end of the volume is the head**, instead of trusting vendor conventions.

Two jobs

- **Trim** non-humeral tails (anatomy-driven thresholds).
- **Flag** internal outliers (slices where SAM grabbed extra tissue).



Trim tails and flag anomalies *before* fitting any geometry.

Feature 1: global humeral-head sphere

What it is

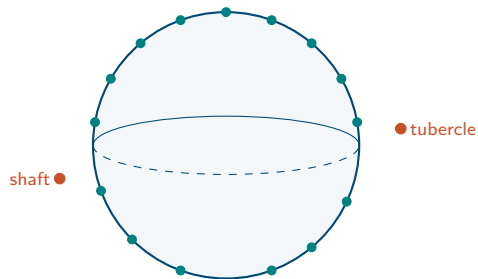
A single sphere fitted, in **millimeters**, to the 3D points on the articular surface.

How

- **Kåsa** algebraic fit — fast, linear, exact in closed form.
- Wrapped in **RANSAC** (2000 iterations, 2 mm inlier distance) to ignore outliers like the shaft or tubercles.

Why it matters

Provides a patient-specific **reference geometry** against which local deviations stand out.



Teal: inliers. Orange: outliers, excluded automatically.

Feature 2: per-slice articular circles

Key idea

On equatorial slices, the articular contour is locally circular. Each fitted circle is an **arc that lies on the cartilage** — not an approximation of it.

Clinical meaning

- A focal **drop in inlier ratio** signals loss of articular continuity — a potential tear.
- Each circle is rendered as a PNG so the radiologist can **verify it slice by slice**.

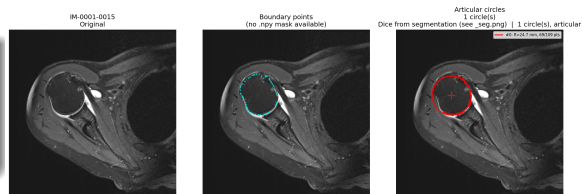


Image → mask → contour → fitted circle.

The geometric trick: the circle as a tendon map

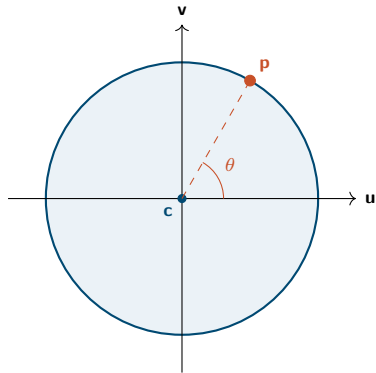
From a circle to one angle

Each per-slice circle has its own $(\mathbf{u}, \mathbf{v})$ axes on the slice plane. Every inlier point on the circle has a single angle θ around the center:

$$\theta = \text{atan2}((\mathbf{p} - \mathbf{c}) \cdot \mathbf{v}, (\mathbf{p} - \mathbf{c}) \cdot \mathbf{u}).$$

Why this is powerful

The same θ corresponds to the **same anatomical direction** on the humeral head — for every patient and every vendor.



Angular sectors → rotator-cuff tendons

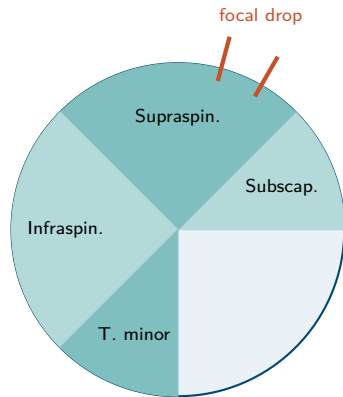
Anatomical mapping

Each tendon inserts in a **predictable angular sector**:

- Subscapularis: $\sim 0-45^\circ$
- Supraspinatus: $\sim 45-135^\circ$
- Infraspinatus: $\sim 135-225^\circ$
- Teres minor: $\sim 225-270^\circ$

Localization rule

A focal drop in inlier ratio inside a sector $\Rightarrow$ a candidate **tear in that tendon**, at the slice where the drop occurs.



The articular circle becomes a 1D *tendon map*.

Multi-vendor robustness and equity of access

Tested across 3 vendors and 3 sites

- **SIEMENS** — Quibdó (Chocó), PD-fs / T1-fs.
- **GE** — Optima MR360, Bogotá, PD-fs.
- **Philips** — Ingenia, SOMA Radiology.

6 datasets, matrices 256×256 to 320×320 .

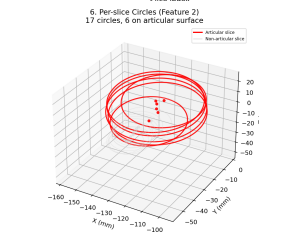
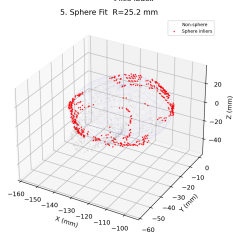
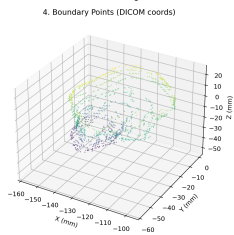
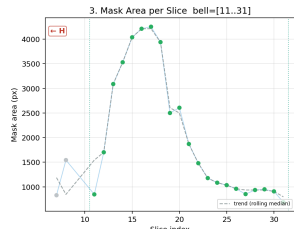
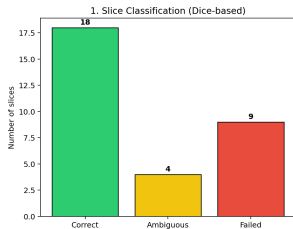
Why this matters in Colombia

Regions like **Chocó** have modern MRI but few subspecialty radiologists. An auditable, interpretable tool can directly support generalist readers — aligned with the UEMS pillar of **impacting** medical practice.

Why it generalizes

Frozen SAM (no overfitting), **geometric** thresholds (mm, % of peak), direction from DICOM headers.

Qualitative results: per-dataset summary



Slice classification, Dice per slice, area curve, 3D boundary, sphere fit, articular circles. **Green** = kept, **gray** = trimmed, **red** = outlier.

Honest limits

- **Small cohort:** 6 datasets — proof of concept.
- **Localization rule is a hypothesis** — needs validation against arthroscopy.
- Automatic seed may fail on atypical positioning.

Why the design still helps

Outputs are **geometric, visualized, and auditable**: the system fails *loudly*, not silently — the property a radiologist needs to trust it.

Three ideas to remember

- A **frozen foundation model** (SAM) plus **classical geometry** produces interpretable shoulder MRI features — with no training.
- Two features — a **global sphere** + **per-slice articular circles** — turn the contour into a **1D tendon map**.
- Robust across SIEMENS, GE and Philips $\Rightarrow$ useful in low-resource radiology settings.

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Thank you — questions welcome.

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